

Muhammad Bilal Nazir

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SUMMARY

Computer Science graduate from FAST NUCES with hands-on experience in full-stack development and AI-powered applications. I enjoy building production-grade software, from scalable software platforms to intelligent systems using GANs, Transformers, and Deep Learning. My work spans software engineering, REST API development, and machine learning, and I am always looking to build applications that are clean, scalable, and impactful.

EDUCATION

Bachelor of Science in Computer Science

2022 – 2026

– Fast National University of Computer and Emerging Sciences, Chiniot-Faisalabad Campus

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript (ES6+), SQL

ML / AI: TensorFlow, PyTorch, Keras, Scikit-learn, NLP, Generative AI, Computer Vision, Transformers, GANs, Reinforcement Learning, IDS, Hugging Face

Web / Full-Stack: HTML5, CSS3, React.js, Node.js, Express.js, FastAPI, REST APIs, MongoDB, MySQL, PostgreSQL, JWT, TailwindCSS

Tools & Other: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Wireshark, Linux, Gradio, Agile

RESEARCH EXPERIENCE

Next-Gen Cyber-Attack Detection | Undergraduate Research Project

- GAN-Powered Framework for Adversarial Attack Generation and IDS Evaluation. Developed a full-stack system using React.js, Node.js, Python (PyTorch/TensorFlow), and PostgreSQL to generate GAN-based adversarial attacks, evaluate ML and signature-based IDS in real time, and enable continuous lifelong learning for IDS improvement. Achieved 30x inference speedup (4.2ms vs 128ms) and 98%+ evasion rates across 7 NIDS architectures on 5 benchmark datasets.

PROJECTS

PhishGuard: AI-Powered Phishing Detection System

- Built a full-stack ML/NLP system detecting phishing URLs and emails with 92-98% URL accuracy and 90-97% email accuracy using XGBoost, SVM, and DistilBERT. Implemented suspicious words highlighting for explainability and real-time confidence scoring. Developed using React, Node.js, and FastAPI with authentication and scan history support.

Stellar Web Manager: Team Collaboration & AI Dashboard

- Developed a SaaS-style platform using React (TypeScript), Tailwind, Node.js, Express, and MongoDB featuring JWT auth, project and task management, analytics dashboards, real-time notifications, and integrated AI chat via OpenAI and Gemini.

Denoising Diffusion Probabilistic Model (DDPM) for Face Image Generation

- Implemented DDPM from scratch in PyTorch trained on 30,000 CelebA-HQ face images using dual T4 GPUs. Built a U-Net with sinusoidal time embeddings, residual blocks, and self-attention at the bottleneck (8.1M parameters). Achieved high-quality face generation evaluated via PSNR and SSIM. Deployed an interactive Gradio application for real-time inference.

ShopHub: MERN E-Commerce Application

- Built a full-stack e-commerce web app using React, Node.js, Express, and MongoDB featuring user authentication, product catalog, shopping cart, and a clean Material-UI interface.

Masked Autoencoder for Self-Supervised Image Learning

- Implemented a ViT-based MAE from scratch in PyTorch for self-supervised learning, trained on TinyImageNet using dual T4 GPUs with patch masking and reconstruction. Built a real-time inference demo using Gradio, demonstrating strong understanding of transformers and representation learning.

Intelligent Traffic Scene Understanding

- Analyzed the UA-DETRAC dataset using HOG, SIFT, SVM, and deep learning models to compare classical versus deep learning approaches, evaluating accuracy and computational efficiency trade-offs. Results demonstrated that deep learning models significantly outperformed classical methods in detection accuracy while maintaining acceptable computational overhead for real-world traffic monitoring applications.

ACTIVITIES

Hackathon Participant | Teaching Assistant | MLSA Member | GDSC Member | Undergraduate Researcher | FYP Team Lead